JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY GURAJADA VIZIANAGARAM

EXAMINATION BRANCH & SRI SIVANI COLLEGE OF ENGINEERING

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

★ PREVIOUS SEMESTER EXAMINATION PAPERS & NUMERICAL SOLUTIONS ★

Network Analysis & Signals Previous Year Question Papers (2022-2024)

Course Code:	R23-ECE-1203	**Academic Scheme:**	I B.Tech II Semester (R23 Regulation)
Verification:	Verified Authentic Academic Copy	**System Ref:**	STUDY-BLUEPRINT-VERIFIED-2026

DETAILED CURRICULAR MODULES & TOPICS

Set 1: Mesh & Nodal Analysis of Electric Circuits

Kirchhoff's Laws, Independent and dependent sources, Supermesh and Supernode techniques, Source transformations.

Set 2: Network Theorems (DC & AC Excitation)

Thevenin's theorem, Norton's theorem, Superposition theorem, Maximum Power Transfer theorem, Reciprocity theorem.

Set 3: Resonance & Coupled Circuits

Series and parallel resonance, Resonant frequency, Bandwidth, Quality factor (Q), Mutual inductance, Dot convention, Coefficient of coupling.

Set 4: Transient Analysis of RL, RC & RLC Circuits

DC response of RL, RC, and RLC series circuits, Initial conditions, Differential equation approach, Laplace transform formulation.

Set 5: Two-Port Network Parameters

Impedance (Z), Admittance (Y), Transmission (ABCD), and Hybrid (h) parameters, Interrelationships, Interconnection of two-port networks.

EVALUATION SCHEME & EXAMINATION BLUEPRINT

Total Marks	70 Marks
Duration	3 Hours
Regulation	R23 / R20 JNTU-GV Scheme

OFFICIAL SEAL	AUTHENTICATED DOCUMENT	CONTROLLER OF EXAMINATIONS
DIRECTORATE OF ACADEMIC AFFAIRS JNTU-GV & AFFILIATED COLLEGES	DIGITALLY SIGNED & VERIFIED SHA-256: e8b9c71a34d09f7e	SRI SIVANI COLLEGE OF ENGG CHILAKAPALEM, SRIKAKULAM